

ENGR 102 Summer Bridge

Day 6 Instructor Key

Multi-Step Calculations, Summaries, and Conditional Output

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Evaluate ceiling-style kit arithmetic.

4 points

```
groups = 7
items_per_group = 5
spares = 4
items = groups * items_per_group + spares
kits = (items + 11) // 12
leftover_space = kits * 12 - items
print(kits, leftover_space)
```

Show the item total, adjusted numerator, quotient, and exact output.

Answer and explanation

items is $7 * 5 + 4 = 39$. Adding 11 before floor division gives $50 // 12 = 4$ kits. Four kits hold 48 items, leaving 9 unused spaces. Exact output: 4 9.

Question 2. Trace a numeric summary and formatting boundary.

4 points

```
values = [4.5, 7.0, 2.5, 6.0]
total = sum(values)
average = total / len(values)
print(min(values), max(values), f"{average:.2f}")
```

Compute every summary and give exact output, including the formatted average.

Answer and explanation

The total is 20.0 across four values, so average is 5.0. The minimum is 2.5 and maximum is 7.0. Two-decimal formatting produces exact output 2.5 7.0 5.00.

Question 3. Combine arithmetic with a conditional label.

4 points

```
mass = 18.0
purity = 84.5
cost = 12 + 1.8 * mass
hold = purity < 85 or mass > 25

if hold:
    label = "hold"
else:
    label = "accepted"

print(f"{label}: {cost:.2f}")
```

Show the cost calculation and each Boolean component before giving exact output.

Answer and explanation

The cost is $12 + 1.8 * 18 = 44.4$. $\text{purity} < 85$ is True and $\text{mass} > 25$ is False, so the [or](#) result hold is True. Exact output: hold: 44.40.

Question 4. Program construction**8 points**

Write one complete Python program that satisfies every requirement.

- Read length, width, and height as floats.
- Print invalid if any dimension is not positive.
- Otherwise compute volume and rectangular-prism surface area.
- Classify oversize when volume is above 500 or the largest dimension is above 12; otherwise classify standard.
- Print Volume and Surface area to two decimals, followed by Class: and the label.

Answer and explanation

```
length = float(input())
width = float(input())
height = float(input())

if length <= 0 or width <= 0 or height <= 0:
    print("invalid")
else:
    volume = length * width * height
    surface_area = 2 * (length * width + length * height + width * height)
    if volume > 500 or max(length, width, height) > 12:
        label = "oversize"
    else:
        label = "standard"
    print(f"Volume: {volume:.2f}")
    print(f"Surface area: {surface_area:.2f}")
    print(f"Class: {label}")
```

The validity branch protects all formulas. Surface area keeps the three face-pair terms visible before doubling their sum. For 10, 5, and 4, volume is 200 and surface area is 220, so the label is standard. For 13, 5, and 8, volume is 520 and the largest dimension is 13, so both independent checks support oversize.

Scoring guidance

1 input, 1 validity, 2 volume/surface formulas, 2 compound classification, 1 exact precision/output, 1 boundary test.